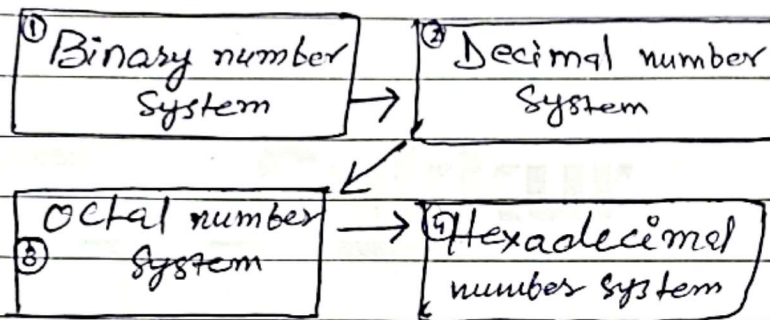


Student's Name: _____ Number System _____ Date: _____
 Class: _____ CMS ID: _____ Course: _____
 Instructor: _____ No. _____

Digital computer, there are various types of number system.



Binary Number System:- In this system carries only two digits either 0 or 1.

Each digit known as a bit. Four bit (1101) is known as nibble and collection of eight bits (11001010) is known as byte.

The location of digit in a binary number represents a specific power of the base (2) of the number system.

$$d \cdot 2^n$$

n is an integer, d is one of the binary digits (or bits) 0 or 1

Ex: $27 = 16 + 8 + 2 + 1$
 $= 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 11011$

$11011 \Rightarrow 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$
 $16 + 8 + 0 + 2 + 1 = 27$

Ex: $1101.011 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} = 8 + 4 + 0 + 1 + 0.25 + 0.125 = 10.375$

Binary to decimal

$$1010 \Rightarrow 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 2^3 + 0 + 2 + 0 = 10$$

$\downarrow \downarrow \downarrow \downarrow$
 $2^3 \ 2^2 \ 2^1 \ 2^0$

$$10010 = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$$

$\downarrow \downarrow \downarrow \downarrow \downarrow$
 $2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$$= 2^4 + 2^1 = 16 + 2 = 18$$

Ex: $1100.101 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3}$

$$2^3 \ 2^2 \ 2^1 \ 2^0 \ 2^{-1} \ 2^{-2} \ 2^{-3} = 2^3 + 2^2 + 2^{-1} + 2^{-3}$$
$$= 8 + 4 + \frac{1}{2} + \frac{1}{8}$$
$$= 12 + 0.5 + 0.125$$
$$= 12.625$$

Decimal to binary:

2	27
2	13 - 1
2	6 - 0
2	3 - 0
	1 - 1

11001

2	25
2	12 - 1
2	6 - 0
2	3 - 0
	1 - 1

$$25 = (11001)_2$$

Ex $12.625 = 12 + 0.625$

$$\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 - 0 \\ \hline 2 & 3 - 0 \\ \hline & 1 - 1 \end{array}$$

1100

$$0.625 \times 2 = 1.25 = 1 + 0.25$$

$$0.25 \times 2 = 0.5 = 0 + 0.5$$

$$0.5 \times 2 = 1 = 1$$

.101

$$(1100.101)_2 = 12.625$$

Number System:

A number system is defined as a system of writing to express numbers. A number system is a mathematical system with base n , where n represents total numbers present in that system.

Eg: In decimal number system, the number of digits is 10

Radix Point:

Radix point is a base point used to separate the integer part and the fractional part of a number.

Base or radix:

The number of symbols used in a number system is called base or radix of the number system.

Eg: In decimal number system – Base is 10

In Binary number system – Base is 2

In Hexadecimal number system – Base is 16 etc.

Most significant bit (MSB):

The left most bit of a binary number which has the highest place value is called most significant bit.

Least significant bit (LSB):

The right most bit of a binary number which has the least place value is called least significant bit.

Bit:

A binary digit is called a bit.

Example: 0 or 1

Nibble:

A group of four bits is called a Nibble.

Example: 0110, 1110 etc.

Byte:

A group of eight bits is called a byte.

Example: 01101101, 11010011 etc.

Types of number systems

- Binary number system
- Octal number system
- Decimal number system
- Hexadecimal number system

Binary number system:

The base of binary number system is 2. The digits used are 0 and 1.

Example: $11001_{(2)}$, $0101_{(2)}$

Octal number system:

The base of octal number system is 8.

The digits used are 0, 1, 2, 3, 4, 5, 6 and 7.

Example: $157_{(8)}$, $2312_{(8)}$, $110_{(8)}$

Table representing decimal, Octal and binary number

Decimal Number	Octal Number	Binary Number
0	0	000
1	1	001
2	2	010
3	3	011
4	4	100
5	5	101
6	6	110
7	7	111

Decimal number system:

The base of decimal number system is 10.

The digits used are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

Example: $0157_{(10)}$, $6587_{(10)}$, $0110_{(10)}$

Hexadecimal number system:

The base of hexadecimal number system is 16.

The digits used are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F.

Example: $0157_{(16)}$, $2312_{(16)}$, $0110_{(16)}$, $ABC5_{(16)}$, $BDEF_{(16)}$

Table representing decimal, hexadecimal and binary number

Decimal Number	Hexadecimal Number	Binary Number
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Binary Arithmetic

Binary arithmetic includes the basic arithmetic operations of **addition, subtraction, multiplication and division**

Binary Addition: The rules to perform binary addition are

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ with a carry } 1$$

Binary Subtraction: The rules to perform binary subtraction are

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ with a borrow } 1$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

Binary Multiplication: The rules to perform binary multiplication are

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Binary Division: The rules to perform binary division are

$$0 \div 0 = \text{not defined}$$

$$0 \div 1 = 0$$

$$1 \div 0 = \text{not defined}$$

$$1 \div 1 = 1$$

1's Complement of binary numbers:

The 1's complement of a binary number is obtained by changing each 0 to 1 and 1 to 0.

Eg: The 1's complement of $(101101)_2$ is $(010010)_2$

2'S Complement of binary numbers: The 2'S complement of a binary number is obtained by adding 1 to the 1's complement of the number.

Eg: The 2's complement of $(101101)_2$ is $(010010)_2 + 1 = (010011)_2$

Note: The 1'S complement and the 2'S complement of a binary number help to do the subtraction by the method of addition.

BINARY CODES

Digital Codes or Binary Codes:

A group of binary digits is known as a binary code or a digital code.

Numeric Code:

Digital code represented by the number is known as a numeric code.

Classification:

The codes are broadly classified into

1. Weighted code
2. Non-weighted code
3. Reflective code
4. Sequential code
5. Error detecting and correcting codes